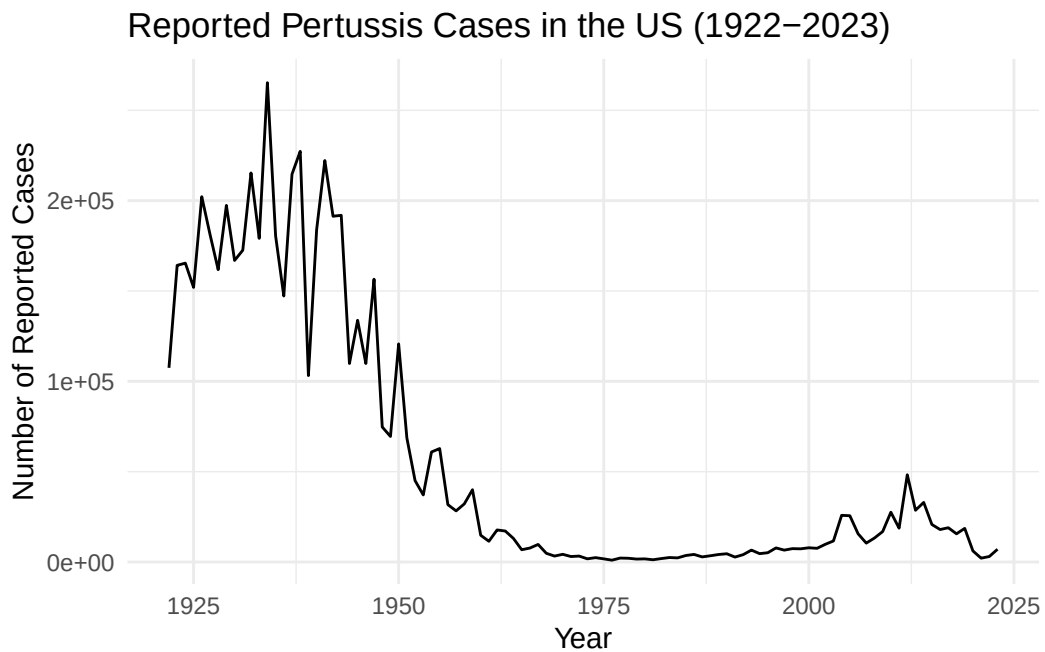


# Pertussis Mini Project

Mari Williams (A15858833)

###Q1

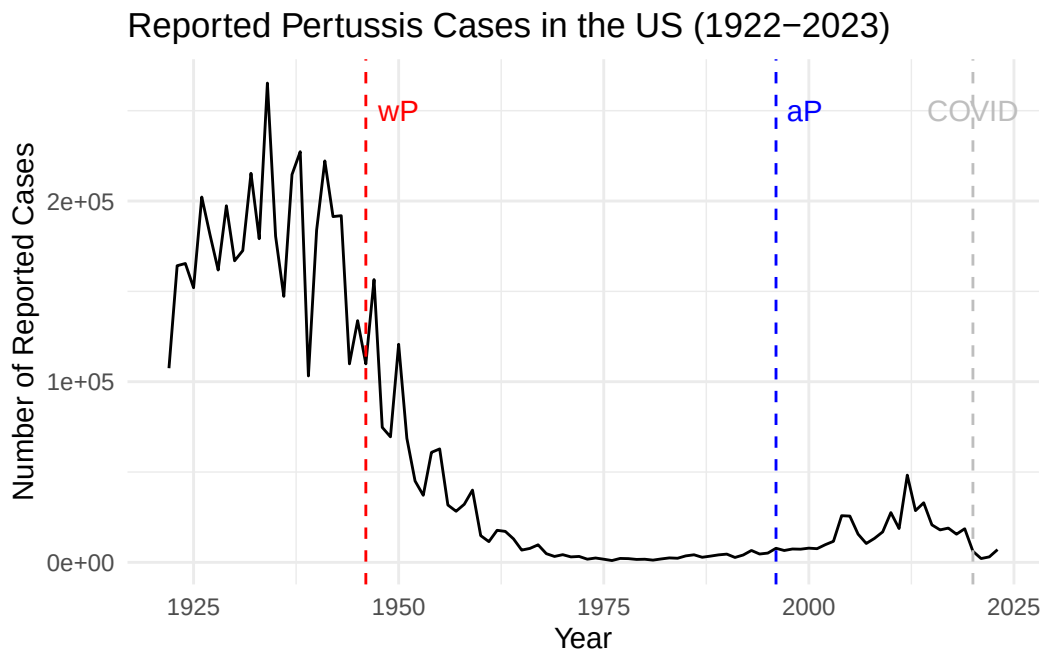
```
library(ggplot2)
ggplot(cdc) +
  geom_line(aes(x = Year, y = No..Reported.Pertussis.Cases)) +
  labs(title = "Reported Pertussis Cases in the US (1922-2023)",
       x = "Year",
       y = "Number of Reported Cases") +
  theme_minimal()
```



###Q2

```
ggplot(cdc) +
  geom_line(aes(x = Year, y = No..Reported.Pertussis.Cases)) +
  labs(title = "Reported Pertussis Cases in the US (1922-2023)",
       x = "Year",
       y = "Number of Reported Cases") +
  theme_minimal() + geom_vline(xintercept = 1946, linetype="dashed",
                              color = "red", size=.5) +
  annotate("text", x = 1950, y = 250000, label = " wP ", color = "red") + geom_vline(xintercept = 1999, linetype="dashed",
                              color = "blue", size=.5) +
  annotate("text", x = 2000, y = 250000, label = "aP ", color = "blue") + geom_vline(xintercept = 2020, linetype="dashed",
                              color = "gray", size=.5) +
  annotate("text", x = 2020, y = 250000, label = " COVID ", color = "gray")
```

Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.  
 i Please use `linewidth` instead.



###Q3 There is gradual increase in pertussis cases after the aP introduction, followed by a few spikes. This is likely due to a growing anti-vaccination movement in the US. It is also due to the fact that aP vaccines do not provide as long-lasting immunity as wP vaccines. The decrease in cases during the COVID-19 pandemic is likely due to social distancing and mask-wearing measures that were implemented to reduce the spread of COVID-19, which also reduced the spread of pertussis.

```
(library(jsonlite))
```

Warning: package 'jsonlite' was built under R version 4.3.3

```
[1] "jsonlite" "ggplot2" "stats" "graphics" "grDevices" "utils"
[7] "datasets" "methods" "base"
```

```
subject <- read_json("https://www.cmi-pb.org/api/subject", simplifyVector = TRUE)
```

```
head(subject, 3)
```

|   | subject_id | infancy_vac | biological_sex | ethnicity              | race  |
|---|------------|-------------|----------------|------------------------|-------|
| 1 | 1          | wP          | Female         | Not Hispanic or Latino | White |
| 2 | 2          | wP          | Female         | Not Hispanic or Latino | White |
| 3 | 3          | wP          | Female         | Unknown                | White |

|   | year_of_birth | date_of_boost | dataset      |
|---|---------------|---------------|--------------|
| 1 | 1986-01-01    | 2016-09-12    | 2020_dataset |
| 2 | 1968-01-01    | 2019-01-28    | 2020_dataset |
| 3 | 1983-01-01    | 2016-10-10    | 2020_dataset |

##Q4, Q5, Q6

```
table(subject$infancy_vac)
```

```
aP wP
87 85
```

```
table(subject$biological_sex)
```

```
Female    Male
   112     60
```

```
table(subject$biological_sex, subject$race)
```

|        | American Indian/Alaska Native | Asian | Black or African American |
|--------|-------------------------------|-------|---------------------------|
| Female | 0                             | 32    | 2                         |
| Male   | 1                             | 12    | 3                         |

|        | More Than One Race | Native Hawaiian or Other Pacific Islander |
|--------|--------------------|-------------------------------------------|
| Female | 15                 | 1                                         |
| Male   | 4                  | 1                                         |

|        | Unknown or Not Reported | White |
|--------|-------------------------|-------|
| Female | 14                      | 48    |
| Male   | 7                       | 32    |

###Q7 and Q8

```
library(lubridate)
```

Warning: package 'lubridate' was built under R version 4.3.3

Attaching package: 'lubridate'

The following objects are masked from 'package:base':

date, intersect, setdiff, union

```
for (i in 1:(length(subject$year_of_birth))) {
  subject$age_years[i] <- as.numeric(time_length( today() - ymd(subject$year_of_birth[i]),
})
```

```
library(dplyr)
```

Warning: package 'dplyr' was built under R version 4.3.3

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
ap <- subject %>% filter(infancy_vac == "aP")
mean(ap$age_years, na.rm = TRUE)
```

```
[1] 27.82827
```

```
wp <- subject %>% filter(infancy_vac == "wP")
mean(wp$age_years, na.rm = TRUE)
```

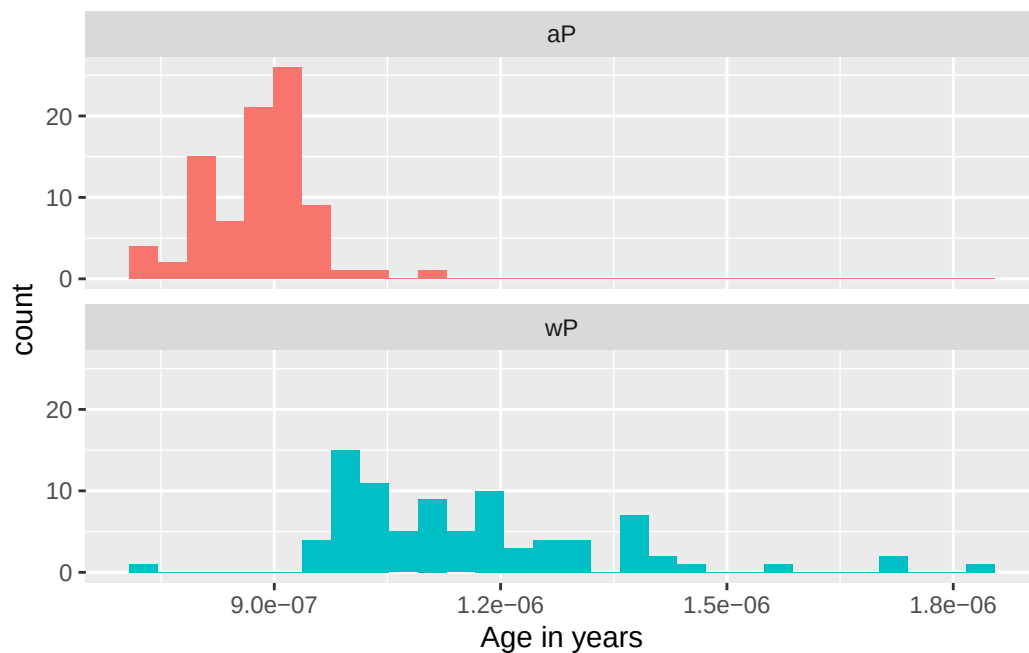
```
[1] 36.57897
```

```
int <- ymd(subject$date_of_boost) - ymd(subject$year_of_birth)
age_at_boost <- time_length(int, "year")
head(age_at_boost)
```

```
[1] 30.69678 51.07461 33.77413 28.65982 25.65914 28.77481
```

```
ggplot(subject) +
  aes(time_length(age_years, "year"),
       fill=as.factor(infancy_vac)) +
  geom_histogram(show.legend=FALSE) +
  facet_wrap(vars(infancy_vac), nrow=2) +
  xlab("Age in years")
```

`stat\_bin()` using `bins = 30`. Pick better value `binwidth`.



Yes, wP group is older and more spread out

```
specimen <- read_json("https://www.cmi-pb.org/api/specimen", simplifyVector = TRUE)
titer <- read_json("https://www.cmi-pb.org/api/plasma_ab_titer", simplifyVector = TRUE)
head(specimen)
```

|   | specimen_id | subject_id | actual_day_relative_to_boost |  |
|---|-------------|------------|------------------------------|--|
| 1 | 1           | 1          | -3                           |  |
| 2 | 2           | 1          | 1                            |  |
| 3 | 3           | 1          | 3                            |  |
| 4 | 4           | 1          | 7                            |  |
| 5 | 5           | 1          | 11                           |  |
| 6 | 6           | 1          | 32                           |  |

|   | planned_day_relative_to_boost | specimen_type | visit |
|---|-------------------------------|---------------|-------|
| 1 | 0                             | Blood         | 1     |
| 2 | 1                             | Blood         | 2     |
| 3 | 3                             | Blood         | 3     |
| 4 | 7                             | Blood         | 4     |
| 5 | 14                            | Blood         | 5     |
| 6 | 30                            | Blood         | 6     |

##Q6

##Q7

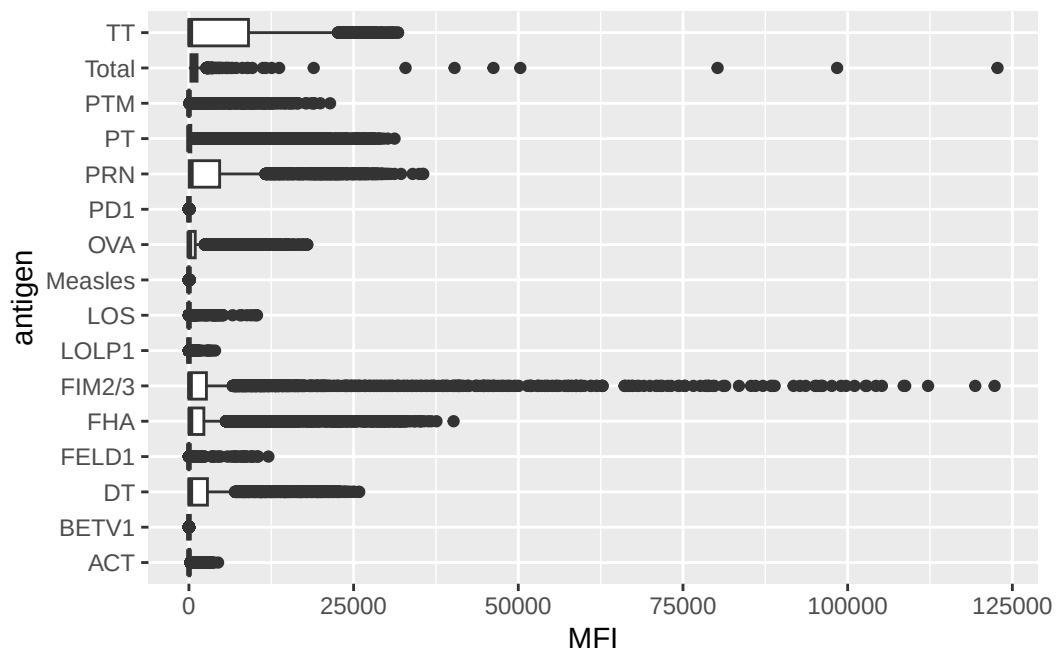
```
unique(titer$antigen)
```

```
[1] "Total"  "PT"      "PRN"      "FHA"      "ACT"      "LOS"      "FELD1"  
[8] "BETV1"  "LOLP1"    "Measles"  "PTM"      "FIM2/3"   "TT"       "DT"  
[15] "OVA"     "PD1"
```

###Q8

```
ggplot(titer)+ aes(MFI, antigen) +  
  geom_boxplot()
```

Warning: Removed 1 row containing non-finite outside the scale range (`stat\_boxplot()`).



###Q9 and Q10

```
library(dplyr)  
meta <- inner_join(specimen, subject)
```

Joining with `by = join\_by(subject\_id)`

```
head(meta)
```

|   | specimen_id | subject_id | actual_day_relative_to_boost |
|---|-------------|------------|------------------------------|
| 1 | 1           | 1          | -3                           |
| 2 | 2           | 1          | 1                            |
| 3 | 3           | 1          | 3                            |
| 4 | 4           | 1          | 7                            |
| 5 | 5           | 1          | 11                           |
| 6 | 6           | 1          | 32                           |

|   | planned_day_relative_to_boost | specimen_type | visit | infancy_vac | biological_sex |
|---|-------------------------------|---------------|-------|-------------|----------------|
| 1 | 0                             | Blood         | 1     | wP          | Female         |
| 2 | 1                             | Blood         | 2     | wP          | Female         |
| 3 | 3                             | Blood         | 3     | wP          | Female         |
| 4 | 7                             | Blood         | 4     | wP          | Female         |
| 5 | 14                            | Blood         | 5     | wP          | Female         |
| 6 | 30                            | Blood         | 6     | wP          | Female         |

|   | ethnicity              | race  | year_of_birth | date_of_boost | dataset      |
|---|------------------------|-------|---------------|---------------|--------------|
| 1 | Not Hispanic or Latino | White | 1986-01-01    | 2016-09-12    | 2020_dataset |
| 2 | Not Hispanic or Latino | White | 1986-01-01    | 2016-09-12    | 2020_dataset |
| 3 | Not Hispanic or Latino | White | 1986-01-01    | 2016-09-12    | 2020_dataset |
| 4 | Not Hispanic or Latino | White | 1986-01-01    | 2016-09-12    | 2020_dataset |
| 5 | Not Hispanic or Latino | White | 1986-01-01    | 2016-09-12    | 2020_dataset |
| 6 | Not Hispanic or Latino | White | 1986-01-01    | 2016-09-12    | 2020_dataset |

|   | age_years |
|---|-----------|
| 1 | 39.93155  |
| 2 | 39.93155  |
| 3 | 39.93155  |
| 4 | 39.93155  |
| 5 | 39.93155  |
| 6 | 39.93155  |

```
dim(meta)
```

```
[1] 1503  14
```

```
abdata<- inner_join(meta, titer)
```

Joining with `by = join\_by(specimen\_id)`



```
head(abdata)
```

|   | specimen_id                   | subject_id    | actual_day_relative_to_boost |               |                |                |       |
|---|-------------------------------|---------------|------------------------------|---------------|----------------|----------------|-------|
| 1 | 1                             | 1             | -3                           |               |                |                |       |
| 2 | 1                             | 1             | -3                           |               |                |                |       |
| 3 | 1                             | 1             | -3                           |               |                |                |       |
| 4 | 1                             | 1             | -3                           |               |                |                |       |
| 5 | 1                             | 1             | -3                           |               |                |                |       |
| 6 | 1                             | 1             | -3                           |               |                |                |       |
|   | planned_day_relative_to_boost | specimen_type | visit                        | infancy_vac   | biological_sex |                |       |
| 1 | 0                             | Blood         | 1                            | wP            | Female         |                |       |
| 2 | 0                             | Blood         | 1                            | wP            | Female         |                |       |
| 3 | 0                             | Blood         | 1                            | wP            | Female         |                |       |
| 4 | 0                             | Blood         | 1                            | wP            | Female         |                |       |
| 5 | 0                             | Blood         | 1                            | wP            | Female         |                |       |
| 6 | 0                             | Blood         | 1                            | wP            | Female         |                |       |
|   | ethnicity                     | race          | year_of_birth                | date_of_boost | dataset        |                |       |
| 1 | Not Hispanic or Latino        | White         | 1986-01-01                   | 2016-09-12    | 2020_dataset   |                |       |
| 2 | Not Hispanic or Latino        | White         | 1986-01-01                   | 2016-09-12    | 2020_dataset   |                |       |
| 3 | Not Hispanic or Latino        | White         | 1986-01-01                   | 2016-09-12    | 2020_dataset   |                |       |
| 4 | Not Hispanic or Latino        | White         | 1986-01-01                   | 2016-09-12    | 2020_dataset   |                |       |
| 5 | Not Hispanic or Latino        | White         | 1986-01-01                   | 2016-09-12    | 2020_dataset   |                |       |
| 6 | Not Hispanic or Latino        | White         | 1986-01-01                   | 2016-09-12    | 2020_dataset   |                |       |
|   | age_years                     | isotype       | is_antigen_specific          | antigen       | MFI            | MFI_normalised | unit  |
| 1 | 39.93155                      | IgE           | FALSE                        | Total         | 1110.21154     | 2.493425       | UG/ML |
| 2 | 39.93155                      | IgE           | FALSE                        | Total         | 2708.91616     | 2.493425       | IU/ML |
| 3 | 39.93155                      | IgG           | TRUE                         | PT            | 68.56614       | 3.736992       | IU/ML |
| 4 | 39.93155                      | IgG           | TRUE                         | PRN           | 332.12718      | 2.602350       | IU/ML |
| 5 | 39.93155                      | IgG           | TRUE                         | FHA           | 1887.12263     | 34.050956      | IU/ML |
| 6 | 39.93155                      | IgE           | TRUE                         | ACT           | 0.10000        | 1.000000       | IU/ML |
|   | lower_limit_of_detection      |               |                              |               |                |                |       |
| 1 | 2.096133                      |               |                              |               |                |                |       |
| 2 | 29.170000                     |               |                              |               |                |                |       |
| 3 | 0.530000                      |               |                              |               |                |                |       |
| 4 | 6.205949                      |               |                              |               |                |                |       |
| 5 | 4.679535                      |               |                              |               |                |                |       |
| 6 | 2.816431                      |               |                              |               |                |                |       |

```
dim(abdata)
```

```
[1] 52576    21
```

###Q11

```
table(abdata$isotype)
```

```
  IgE   IgG  IgG1  IgG2  IgG3  IgG4  
6698  5389 10117 10124 10124 10124
```

###Q12

```
table(abdata$dataset)
```

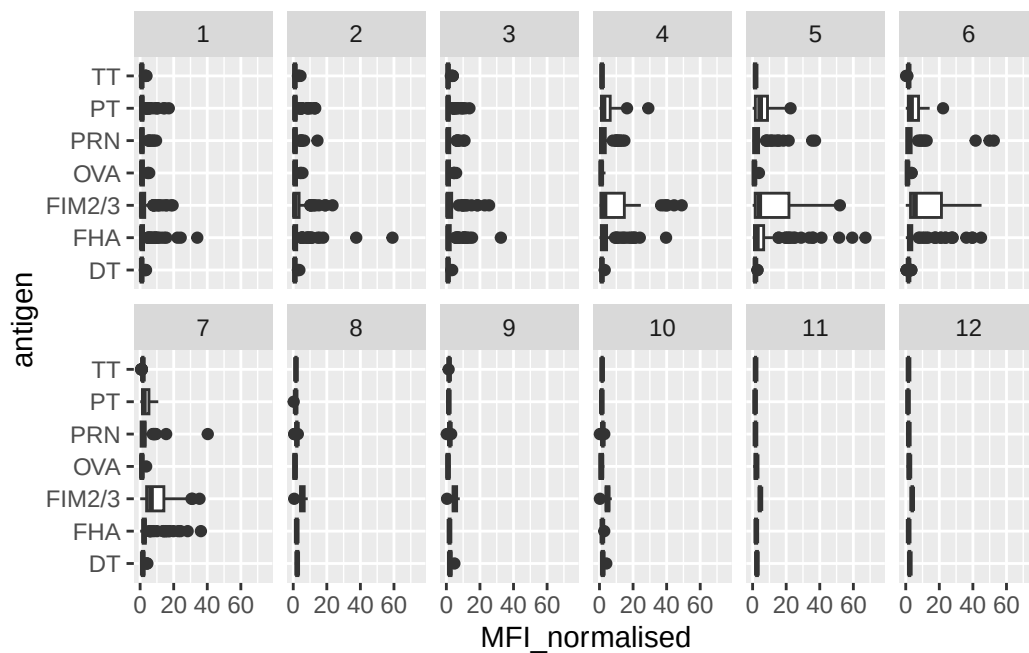
```
2020_dataset 2021_dataset 2022_dataset 2023_dataset  
      31520         8085         7301         5670
```

There's 4 different dataset years. They decrease in size from 2020 to 2023, with 2023 being the smallest.

###Q13

```
igg <- abdata %>% filter(isotype == "IgG")  
ggplot(igg) +  
  aes(x = MFI_normalised, y = antigen) +  
  geom_boxplot() +  
  xlim(0,75) +  
  facet_wrap(vars(visit), nrow=2)
```

Warning: Removed 5 rows containing non-finite outside the scale range (`stat\_boxplot()`).



#### Q14 Time course analysis

```
abdata.tc <- abdata %>% filter(dataset == "2021_dataset")

abdata.tc %>%
  filter(isotype == "IgG", antigen == "PT") %>%
  ggplot() +
    aes(x=planned_day_relative_to_boost,
         y=MFI_normalised,
         col=infancy_vac,
         group=subject_id) +
    geom_point() +
    geom_line() +
    geom_vline(xintercept=0, linetype="dashed") +
    geom_vline(xintercept=14, linetype="dashed") +
    labs(title="2023 dataset IgG PT")
```

2023 dataset IgG PT

